

SCHIZOPHRENIA

& Antipsychotic Medications

Schizophrenia

symptoms
a neurodevelopmental disorder

Antipsychotics

neuroleptics (typical antipsychotics)
atypical (second generation) antipsychotics

Managing Schizophrenia

choosing an antipsychotic
switching antipsychotics

Schizophrenia

- is a neurodevelopmental disorder
- gene x environment interaction
- abnormal development of PFC and hippocampus

First-line treatments

- largely target the dopamine system.

THE SYMPTOMS OF SCHIZOPHRENIA



behavioral excesses

- incoherent speech, loose associations
- delusions
- hallucinations
- odd, bizarre behavior



behavioral deficits

- inappropriate or flat affect
- inattention to self-care
- social withdrawal



cognitive

- resemble PFC dysfunction
- working memory deficits

Schizophrenia is a neurodevelopmental disorder

Gene x environment interactions

- Genetic predisposition
- Synapse structure, function, plasticity
- Environmental stressors
- Perinatal / immunological factors confer risk

alter the developmental trajectory of PFC

- anatomical & functional deficits
- symptoms that resemble abnormal PFC function

Drug therapy is indispensable in treating schizophrenia

- There are two main families of antipsychotic medications
- Typical antipsychotics (neuroleptics; FGA)
- Atypical antipsychotics (SGA)

The primary targets of antipsychotics are

- Dopamine (DA) receptors
- Serotonin (5-HT) receptors

Neuroleptics

Phenothiazine

Butyrophenone

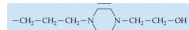
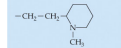
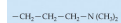
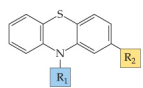
chlorpromazine
(aliphatic)

haloperidol

thioridazine
(piperidine)

fluphenazine
(piperazine)

Phenothiazine nucleus

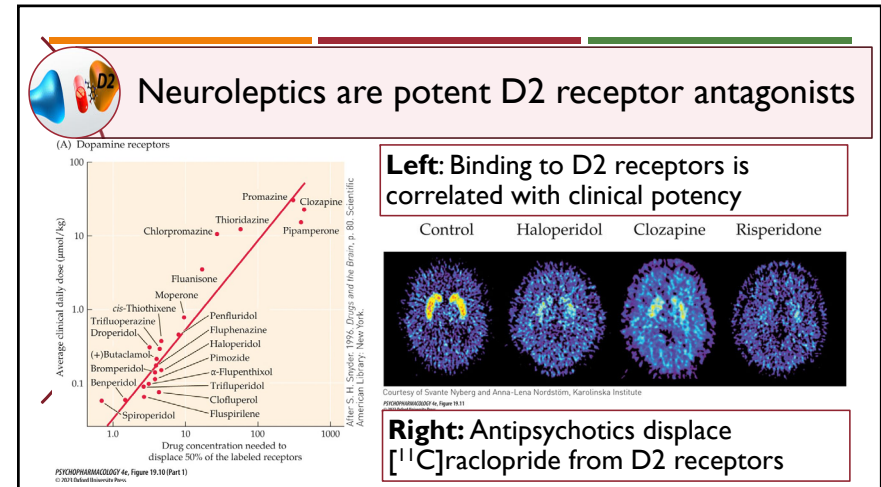
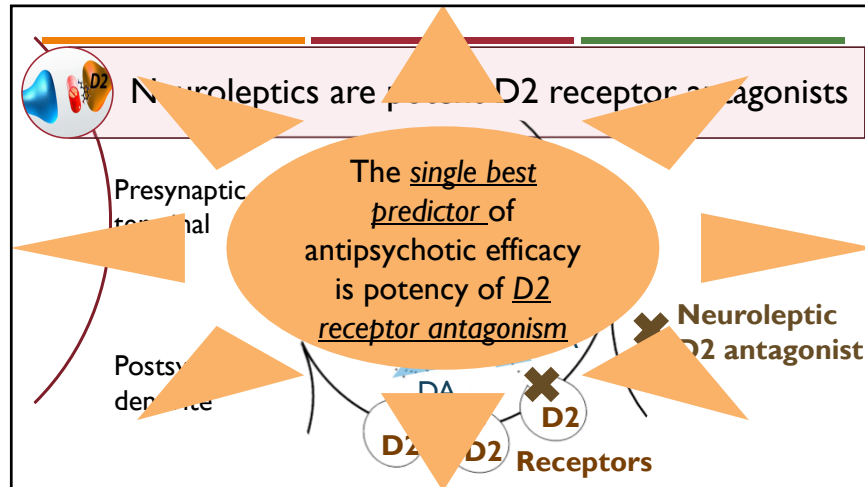


Neuroleptics

are potent D2 receptor antagonists

are effective antipsychotics

have significant side-effects



Neuroleptics are effective antipsychotics

Clinical Efficacy

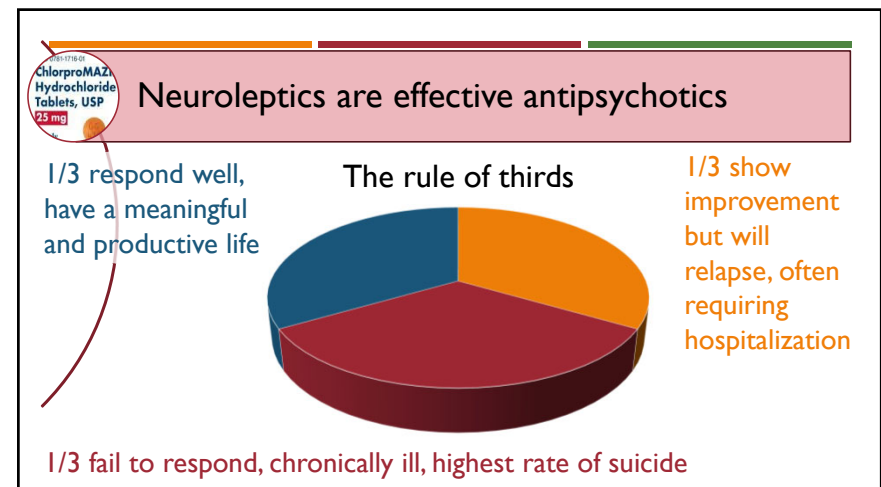
- of neuroleptics has been shown in 100's of double-blind RCTs

Symptoms

- effective in treating the + symptoms
- — and C symptoms are difficult to treat

Maintenance

- After treating acute psychosis, antipsychotic drugs are prescribed for maintenance.
- Discontinuation is often not attempted.



Neuroleptics are potent D2 receptor antagonists

DA cell groups and pathways

Substantia Nigra (A9)	VTA (A10)	VTA (A10)	Hypothalamus (A12)
Nigro-striatal	Meso-limbic	Meso-cortical	Tubero-hypophyseal
• Activation • Motivation • Cognition	• Behavioral arousal • Reward learning	• Attention • Working memory	• Prolactin release

Neuroleptics are potent D2 receptor antagonists

D2 receptors are differentially distributed in the brain

Pathway	D1	D2	Function
Nigro-striatal	+++	+++	EPS side effects
Meso-limbic	+++	+++	Antipsychotic efficacy
Meso-cortical	+	+	Cognition
Tubero-hypophyseal	O	+++	endocrine side effects

Receptor Expression: O None, + Moderate, +++ High

neuroleptics

Neuroleptics have significant side-effects

EPS side effects

Blocking D2 receptors in the nigro-striatal pathway can cause

- dystonia, parkinsonism
- tardive dyskinesia

Endocrine side effects

Blocking D2 receptors in the pituitary increases the release of prolactin

- males: gynecomastia, delayed ejaculation
- females: galactorrhea, amenorrhea, decreased libido

Neuroleptics have significant HAM side-effects

Histamine H1 antagonist

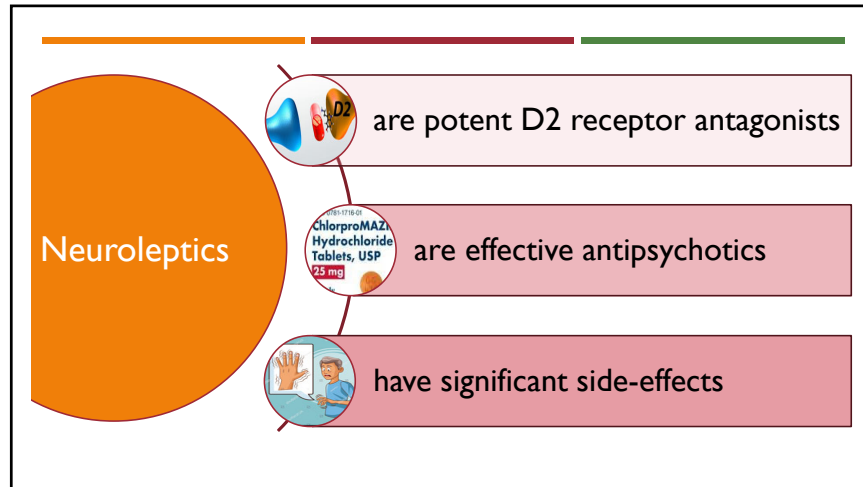
- sedation,
- weight gain

Alpha₁ NE antagonist

- postural hypotension

MACH antagonist

- dry mouth, pupil dilation / blurred vision, cognitive impairment, constipation, tachycardia

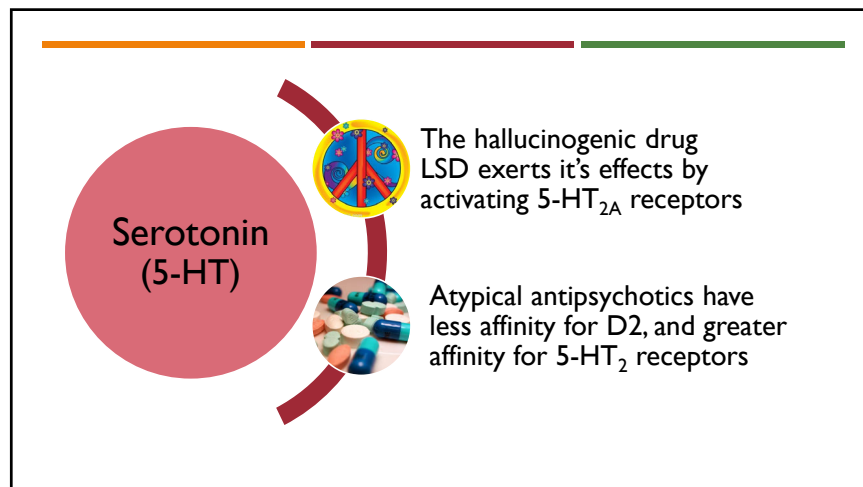


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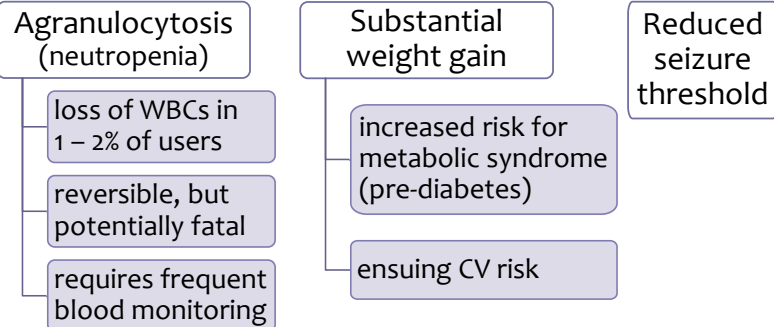
- Dopamine (DA) receptors
- Serotonin (5-HT) receptors



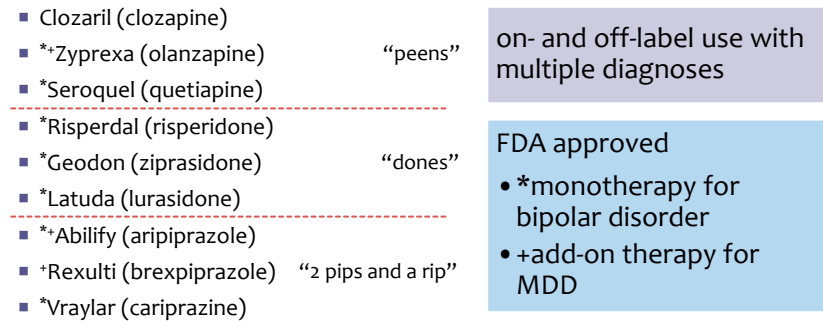
CLOZAPINE HAS SOME ADVANTAGES OVER NEUROLEPTICS

- + Effective in about 1/3 of people non-responsive to neuroleptics
- ↪ EPS (parkinsonian) side-effects are less severe
- ↪ Reduces suicide risk in schizophrenia and schizoaffective disorder

SEVERE SIDE EFFECTS LIMIT THE CLINICAL USE OF CLOZAPINE



“ATYPICAL” ANTIPSYCHOTICS (SGA)



WHAT MAKES AN ANTIPSYCHOTIC “ATYPICAL” ?

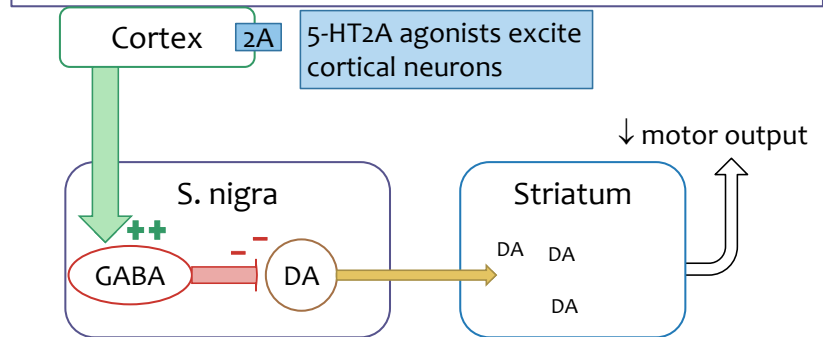
Receptor affinity

- Low affinity **D2 antagonist**
- High affinity **5-HT_{2A} antagonist** (or 5-HT_{1A} partial agonist, or D2 partial agonist)

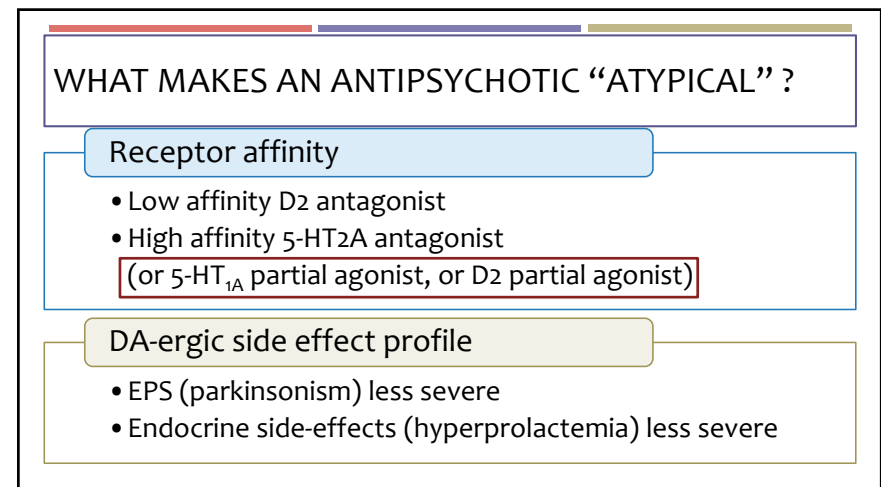
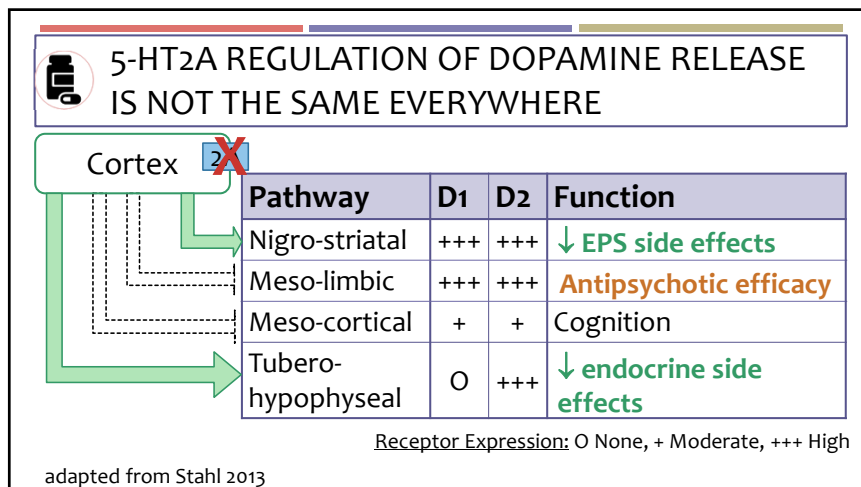
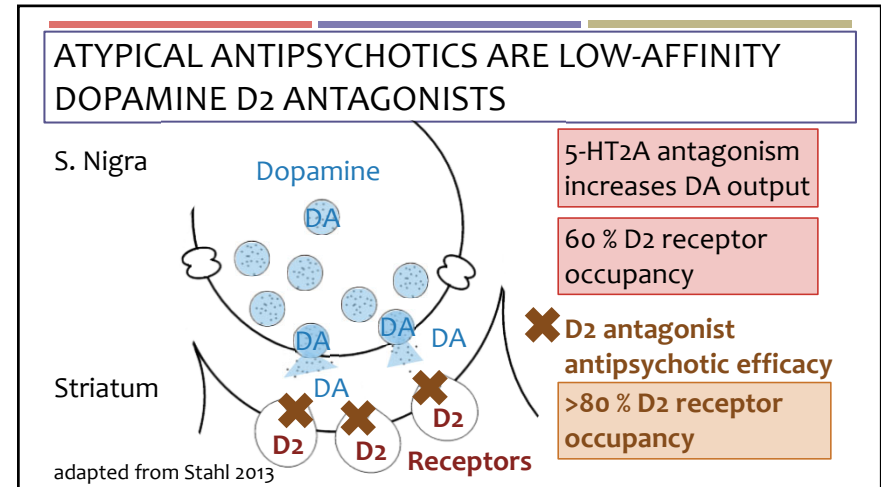
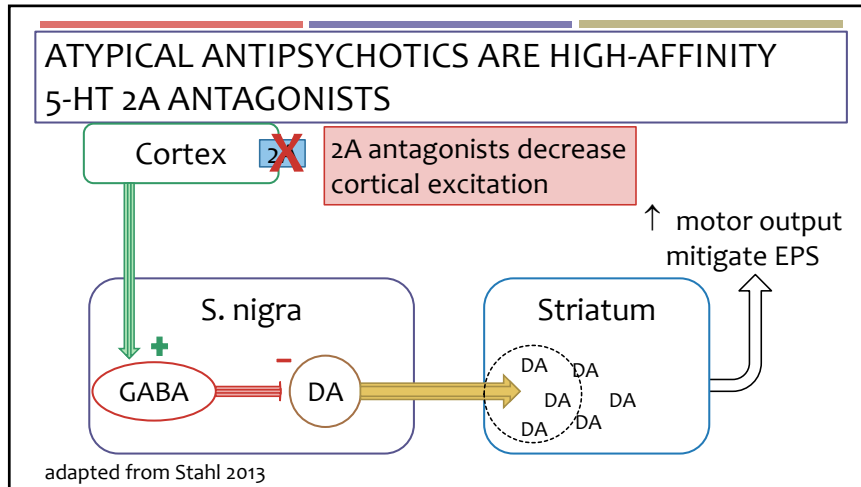
DA-ergic side effect profile

- EPS (parkinsonism) less severe
- Endocrine side-effects (hyperprolactemia) less severe

PHARMACOLOGICAL ACTION OF ATYPICAL ANTIPSYCHOTICS



adapted from Stahl 2013



5-HT_{1A} & D₂ PARTIAL AGONISM CONTRIBUTES TO ANTIPSYCHOTIC & AD EFFICACY

“peens” & “dones”

High affinity 5-HT_{2A} antagonists

- Similar to 5-HT_{2A} antagonism, 5-HT_{1A} *partial agonism* also increase DA release selectively in nigro-striatal pathway and pituitary.
- D₂ *partial agonist* “stabilizes” DA

“2 pips and a rip”

Lower affinity for 5-HT_{2A} receptors

5-HT_{1A} *partial agonist*
D₂ *partial agonist*

5-HT_{1A} & D₂ PARTIAL AGONISM CONTRIBUTES TO ANTIPSYCHOTIC & AD EFFICACY

“peens” & “dones”

5-HT_{1A} *partial agonist*

Depressive episodes in BD
• Seroquel (quetiapine)
• Latuda (lurasidone)

“2 pips and a rip”

D₂ *partial agonist*
5-HT_{1A} *partial agonist*

Adjunct in MDD
• Abilify (aripiprazole)
• Rexulti (brexpiprazole)

GENERAL SIDE EFFECTS OF ATYPICAL ANTIPSYCHOTICS

Sedation

- H₁, ACh, α₁ antagonism
- clozapine (Clozaril)
- olanzapine (Zyprexa)
- quetiapine (Seroquel)

Cardiometabolic

- High
 - clozapine (Clozaril)
 - olanzapine (Zyprexa)
- Moderate
 - risperidone (Risperdal)
 - quetiapine (Seroquel)

THE CARDIOMETABOLIC EFFECTS OF ATYPICAL ANTIPSYCHOTICS

Increased appetite

Cardiovascular events

Weight gain

Diabetes

Elevated triglycerides

Insulin resistance

USE OF ANTIPSYCHOTICS TO MANAGE SCHIZOPHRENIA

Acute psychosis

- Neuroleptics – alone or in combo with BDZ and/or anticholinergic (e.g. benztropine)
- Olanzapine (Zyprexa IM / ODT) or ziprasidone (Geodon for injection)

First episode psychosis

- Atypicals often preferred due to more favorable side-effect profile
- Start with low dose, titrate slowly

CAN'T TOLERATE THE SIDE-EFFECTS? SWITCH

Adverse Effect	1st Alternative	Other	Worst
EPS	Clozaril; Zyprexa	Seroquel; Abilify	Haldol; Thorazine
Sedation	Invega; Fanapt	Abilify; Latuda	Clozaril; Thorazine
Weight gain	Haldol; Geodon	Latuda; Abilify	Zyprexa; Clozaril
Hyper-prolactinemia	Abilify; Seroquel	Saphris; Zyprexa	Invega; Risperdal
Long QT _c	Latuda; Abilify	Invega; Haldol	*Serlect; Geodon

Leucht, S., Cipriani, A., Spinelli, L., Mavridis, D., Orey, D., Richter, F., Samara, M. ... Davis, J.M. (2013). Comparative efficacy and tolerability of 15 antipsychotic drugs in schizophrenia: A multiple-treatments meta-analysis. *Lancet*, 382, 951-962. doi: 10.1016/S0140-6736(13)60733-3

SUMMARY

Atypical

- pharmacological action
- side-effect profile
 - weight gain / type 2 diabetes is problematic

First-line

- Atypical antipsychotics are preferred over neuroleptics
- Choice of antipsychotics is based mostly on PK properties and side-effect tolerability